

Formulae to use

$$X(\omega) = \mathcal{F}\{x(t)\} = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$$
$$x(t) = \mathcal{F}^{-1}\{X(\omega)\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega)e^{j\omega t} d\omega$$
$$\int_{-\infty}^{\infty} e^{-j\alpha t} dt = 2\pi \delta(\alpha)$$

Exercise 1 Show that

$$\mathcal{F}\{e^{j\omega_0 t}\} = 2\pi \delta(\omega - \omega_0)$$

Solution:

$$x(t) = e^{j\omega_0 t}$$
$$\Rightarrow X(\omega) = \int_{-\infty}^{\infty} e^{j\omega_0 t} e^{-j\omega t} dt = \int_{-\infty}^{\infty} e^{-j(\omega - \omega_0)t} dt$$
$$\alpha = \omega - \omega_0 \quad \Rightarrow \quad X(\omega) = 2\pi \delta(\omega - \omega_0)$$

Exercise 2 Determine $\mathcal{F}\{\cos(\omega_0 t)\}$

Solution:

$$\cos(\omega_0 t) = \frac{1}{2} (e^{j\omega_0 t} + e^{-j\omega_0 t})$$
$$\mathcal{F}\{a x(t) + b y(t)\} = aX(\omega) + bY(\omega)$$
$$\Rightarrow \mathcal{F}\{\cos(\omega_0 t)\} = \frac{1}{2} \mathcal{F}\{e^{j\omega_0 t}\} + \frac{1}{2} \mathcal{F}\{e^{-j\omega_0 t}\}$$
$$= \frac{1}{2} (2\pi \delta(\omega - \omega_0)) + \frac{1}{2} (2\pi \delta(\omega + \omega_0))$$
$$\boxed{\mathcal{F}\{\cos(\omega_0 t)\} = \pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]}$$

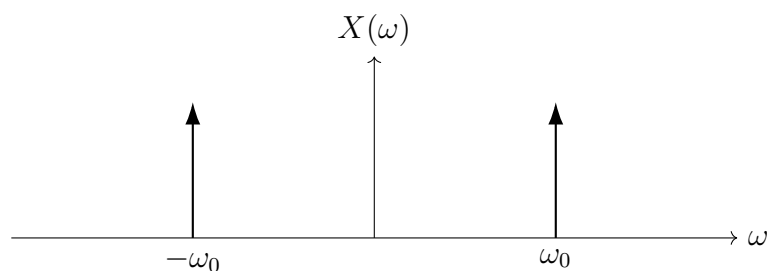


Figure 1: Fourier transform of $\cos(\omega_0 t)$: two spectral lines at $\pm\omega_0$.

Exercise 3 Determine

$$\mathcal{F}\left\{e^{-t^2/\tau^2}\right\}$$

Solution:

$$x(t) = e^{-t^2/\tau^2}$$

$$\Rightarrow X(\omega) = \int_{-\infty}^{\infty} e^{-t^2/\tau^2} e^{-j\omega t} dt = \int_{-\infty}^{\infty} e^{-t^2/\tau^2 - j\omega t} dt$$

$$-\frac{t^2}{\tau^2} - j\omega t = -\frac{1}{\tau^2} \left(t^2 + j\omega\tau^2 t \right)$$

$$t^2 + j\omega\tau^2 t = \left(t + \frac{j\omega\tau^2}{2} \right)^2 - \left(\frac{j\omega\tau^2}{2} \right)^2$$

$$\Rightarrow -\frac{t^2}{\tau^2} - j\omega t = -\frac{1}{\tau^2} \left(t + \frac{j\omega\tau^2}{2} \right)^2 - \frac{\omega^2\tau^2}{4}$$

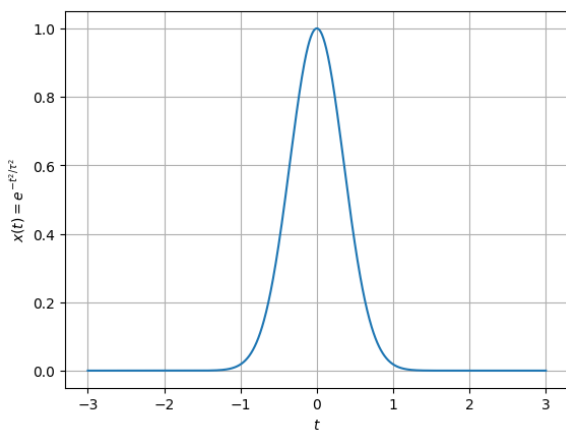
$$\Rightarrow X(\omega) = e^{-\omega^2\tau^2/4} \int_{-\infty}^{\infty} \exp\left[-\frac{1}{\tau^2} \left(t + \frac{j\omega\tau^2}{2} \right)^2 \right] dt$$

$$u = \frac{1}{\tau} \left(t + \frac{j\omega\tau^2}{2} \right) \Rightarrow dt = \tau du$$

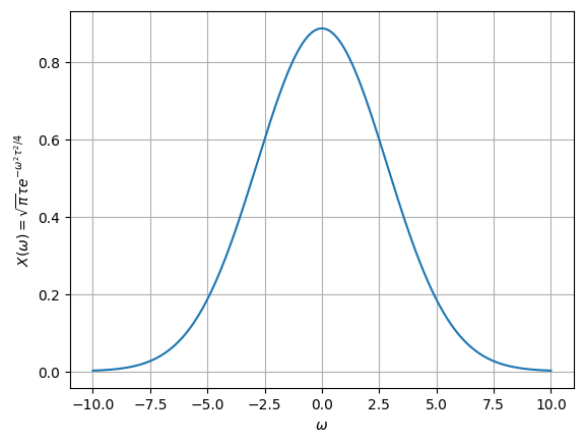
$$\Rightarrow X(\omega) = \tau e^{-\omega^2\tau^2/4} \int_{-\infty}^{\infty} e^{-u^2} du$$

$$\int_{-\infty}^{\infty} e^{-u^2} du = \sqrt{\pi}$$

$$\boxed{\mathcal{F}\left\{e^{-t^2/\tau^2}\right\} = \sqrt{\pi} \tau e^{-\omega^2\tau^2/4}}$$



(a) Time-domain signal $x(t) = e^{-t^2/\tau^2}$.



(b) Fourier transform $X(\omega) = \sqrt{\pi} \tau e^{-\omega^2\tau^2/4}$.

Figure 2: Gaussian signal and its Fourier transform (for $\tau = 1$).

Exercise 4 Determine

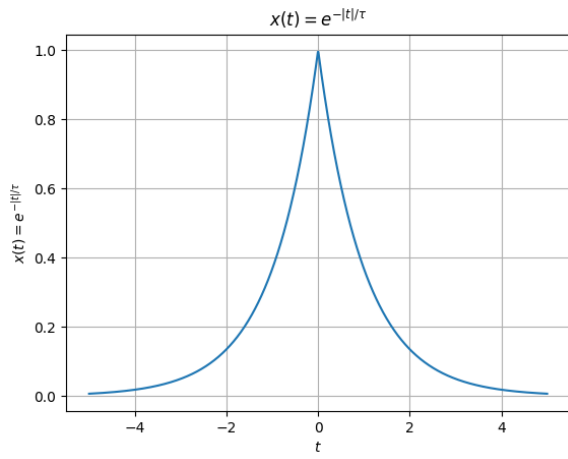
$$\mathcal{F}\{e^{-|t|/\tau}\}$$

Solution:

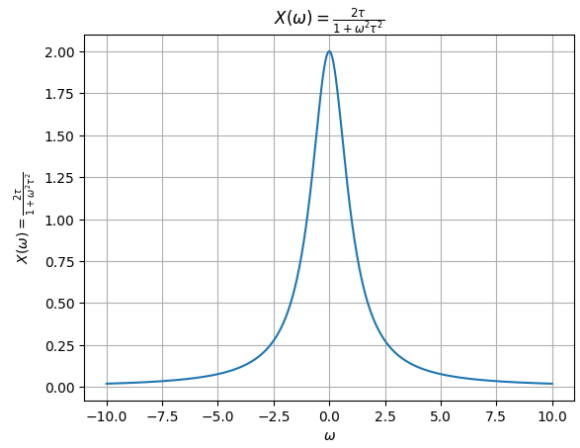
$$\begin{aligned}
 x(t) &= e^{-|t|/\tau} \\
 \Rightarrow X(\omega) &= \int_{-\infty}^{\infty} e^{-|t|/\tau} e^{-j\omega t} dt \\
 &= \int_{-\infty}^0 e^{t/\tau} e^{-j\omega t} dt + \int_0^{\infty} e^{-t/\tau} e^{-j\omega t} dt \\
 &= \int_{-\infty}^0 e^{(1/\tau - j\omega)t} dt + \int_0^{\infty} e^{-(1/\tau + j\omega)t} dt \\
 &= \left[\frac{e^{(1/\tau - j\omega)t}}{1/\tau - j\omega} \right]_{-\infty}^0 + \left[\frac{-e^{-(1/\tau + j\omega)t}}{1/\tau + j\omega} \right]_0^{\infty} \\
 &= \frac{1}{1/\tau - j\omega} + \frac{1}{1/\tau + j\omega} \\
 &= \frac{(1/\tau + j\omega) + (1/\tau - j\omega)}{(1/\tau)^2 + \omega^2} = \frac{2/\tau}{(1/\tau)^2 + \omega^2} \\
 \boxed{\mathcal{F}\{e^{-|t|/\tau}\} &= \frac{2\tau}{1 + \omega^2\tau^2}}
 \end{aligned}$$

Alternative solution

$$\begin{aligned}
 x(t) &= e^{-|t|/\tau} \quad \Rightarrow \quad x(-t) = x(t) \\
 \Rightarrow X(\omega) &= \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt = 2 \int_0^{\infty} x(t) \cos(\omega t) dt \\
 &= 2 \int_0^{\infty} e^{-t/\tau} \cos(\omega t) dt \\
 \cos(\omega t) &= \Re(e^{j\omega t}) \quad \Rightarrow \quad X(\omega) = 2 \Re \left(\int_0^{\infty} e^{-t/\tau} e^{j\omega t} dt \right) \\
 &= 2 \Re \left(\int_0^{\infty} e^{-(1/\tau - j\omega)t} dt \right) \\
 &= 2 \Re \left(\left[\frac{-e^{-(1/\tau - j\omega)t}}{1/\tau - j\omega} \right]_0^{\infty} \right) = 2 \Re \left(\frac{1}{1/\tau - j\omega} \right) \\
 \frac{1}{1/\tau - j\omega} &= \frac{1/\tau + j\omega}{(1/\tau)^2 + \omega^2} \\
 \Rightarrow \Re \left(\frac{1}{1/\tau - j\omega} \right) &= \frac{1/\tau}{(1/\tau)^2 + \omega^2} \\
 \Rightarrow X(\omega) &= 2 \frac{1/\tau}{(1/\tau)^2 + \omega^2} = \frac{2\tau}{1 + \omega^2\tau^2} \\
 \boxed{\mathcal{F}\{e^{-|t|/\tau}\} &= \frac{2\tau}{1 + \omega^2\tau^2}}
 \end{aligned}$$



(a) Time-domain signal $x(t) = e^{-|t|/\tau}$.



(b) Fourier transform $X(\omega) = \frac{2\tau}{1+\omega^2\tau^2}$.

Figure 3: Signal and its Fourier transform (for $\tau = 1$).